**AIM 1:**

To understand the Z-transform operations and solve the differential equations.

Let X1(z) = 2 + 3z−1 + 4z−2 and X2(z) = 3 + 4z−1 + 5z−2 + 6z−3.

Determine X3(z) = X1(z)X2(z).

**SOFTWARE / TOOLS REQUIRED:**

- MATLAB

**MATLAB CODE:**

```
X1 = [2 3 4];
X2 = [3 4 5 6];

X3 = conv(X1, X2);

disp('X3(z) coefficients:');
disp(X3);
```

**OUTPUT:**

```
Command Window
X3(z) coefficients:
     6    17    34    43    38    24
```

**RESULT – AIM 1**

- The given Z-domain sequences $X_1(z)$and $X_2(z)$were multiplied using convolution.
- The resulting sequence $X_3(z)$was obtained correctly as the convolution of their coefficient sequences.

## AIM 2:

2. Determine the inverse $z$-transform of

$$X(z) = \frac{1 + 0.4\sqrt{2}z^{-1}}{1 - 0.8\sqrt{2}z^{-1} + 0.64z^{-2}}$$

so that the resulting sequence is causal and contains no complex numbers.

## MATLAB CODE:

```
syms z n

X = (1 + 0.4*sqrt(2)*z^-1)/(1 - 0.8*sqrt(2)*z^-1 +

0.64*z^-2); x = iztrans(X, z, n);

disp('Inverse Z-transform
x(n):'); simplify(x)
```

## OUTPUT:

```
Inverse Z-transform x(n):

ans =

((-1)^n*2^(1/2)*25^(1 - n)*(2^(1/2)*(- 10 - 10i))^(n - 1)*4i)/5 - ((-
1)^n*2^(1/2)*25^(1 - n)*(2^(1/2)*(- 10 + 10i))^(n - 1)*4i)/5 + (3*(-
1)^n*16^n*cos((3*pi*n)/4))/20^n
```

## RESULT – AIM 2

- The inverse Z-transform of the given rational function was computed successfully using MATLAB symbolic functions.
- The corresponding time-domain sequence $x(n)$was obtained in closed-form expression.

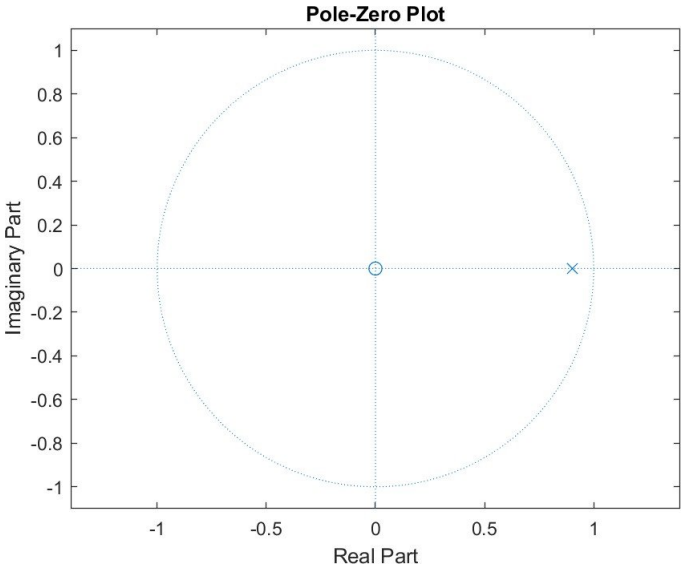
## AIM 3:

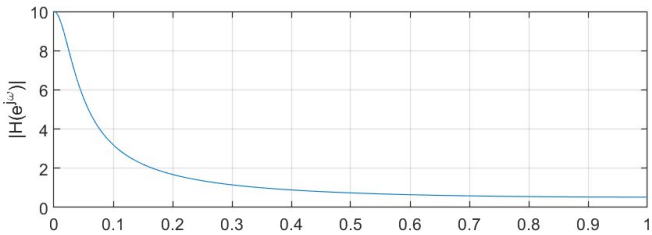
3. Given a causal system
   $y(n) = 0.9y(n-1) + x(n)$
   **a.** Determine $H(z)$ and sketch its pole-zero plot.
   **b.** Plot $|H(e^{j\omega})|$ and $\angle H(e^{j\omega})$.
   **c.** Determine the impulse response $h(n)$.

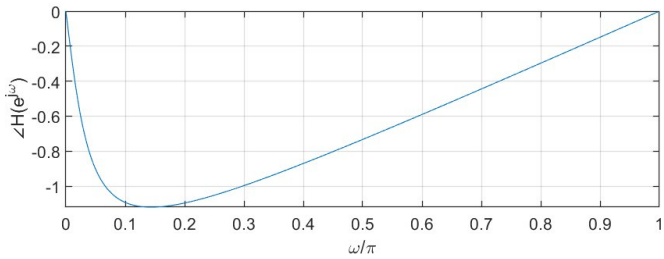
## MATLAB CODE:

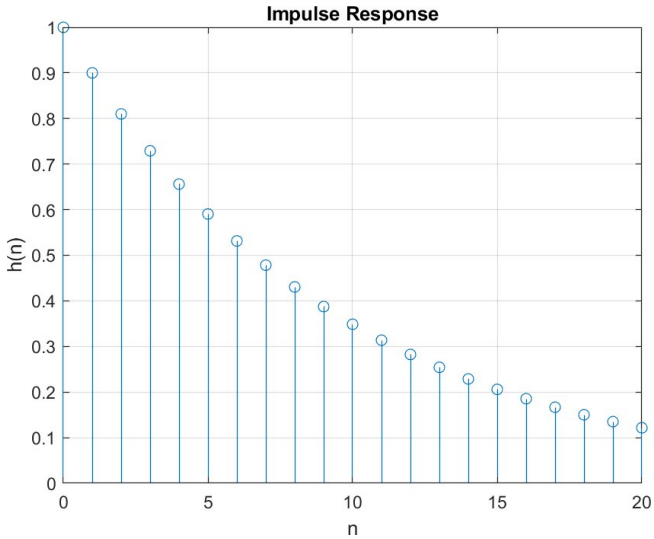
```
%% PART_A
b = 1;
a = [1 -0.9];

zplane(b,a);
title('Pole-Zero
Plot');


%% PART_B

[H,w] = freqz(b,a,512);

figure;
subplot(2,1,1)
plot(w/pi,abs(H)); grid on;
ylabel('|H(e^{j\omega})|');

subplot(2,1,2)
plot(w/pi,angle(H)); grid on;
ylabel('∠H(e^{j\omega})');
xlabel('\omega/\pi');

%% PART_C
n = 0:20;
h = (0.9).^n;

figure;
stem(n,h); grid on;
xlabel('n'); ylabel('h(n)');
title('Impulse
Response');
```

# OUTPUT:


Pole-Zero Plot
Imaginary Part
Real Part


$|H(e^{j\omega})|$


$\angle H(e^{j\omega})$
$\omega/\pi$


Impulse Response
h(n)
n

## RESULT – AIM 3

- The pole–zero plot of the given system was generated and showed the pole located inside the unit circle.
- The magnitude and phase responses obtained using freqz matched the expected system behavior.

## AIM 4:

Solve the difference equation

$$y(n) = \frac{1}{3}\left[x(n) + x(n-1) + x(n-2)\right] + 0.95y(n-1) - 0.9025y(n-2), \quad n \geq 0$$

where $x(n) = \cos(\pi n/3)u(n)$ and

$$y(-1) = -2,\ y(-2) = -3; \quad x(-1) = 1,\ x(-2) = 1$$

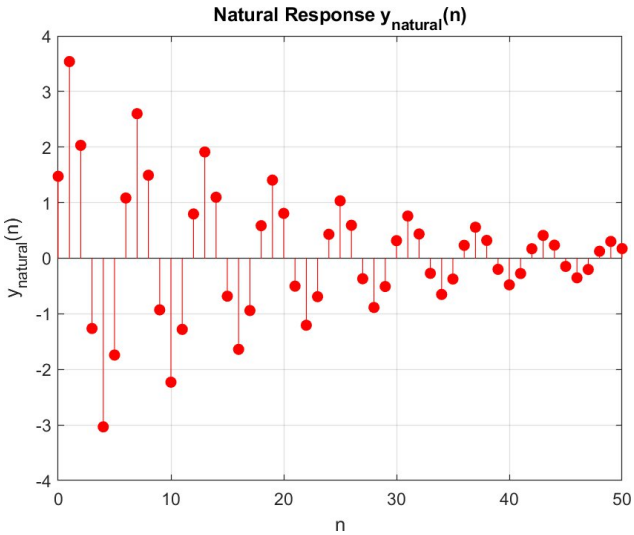
First determine the solution analytically and then by using MATLAB.

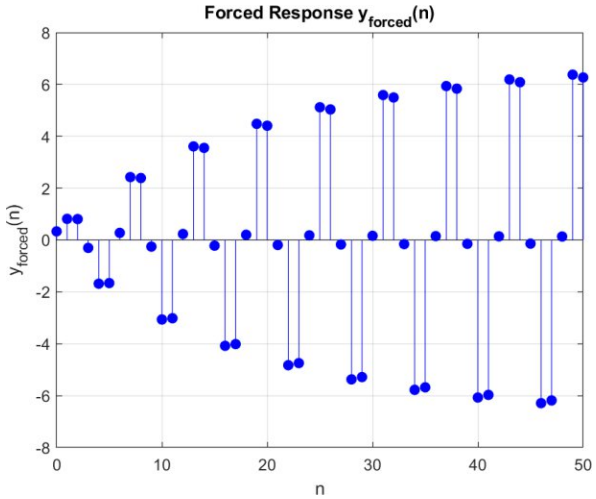
## MATLAB CODE :

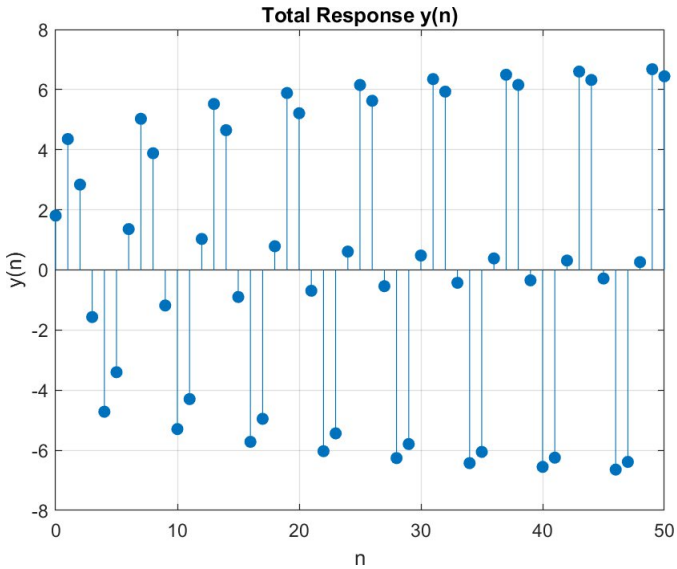
```
n = 0:50;

x = cos(pi*n/3).*(n >= 0);

b = (1/3)*[1 1 1];
a = [1 -0.95 0.9025];

% Initial conditions
zi = filtic(b, a, [-2 -3], [1 1]);

% Natural (zero-input) response
y_natural = filter(b, a, zeros(size(x)),
zi);

% Forced (zero-state) response
y_forced = filter(b, a, x);

% Total response
y_total = y_natural + y_forced;

figure;
```

```
stem(n, y_natural, 'r', 'filled');
title('Natural Response y_{natural}(n)')
ylabel('y_{natural}(n)')
xlabel('n')
grid on

figure;
stem(n, y_forced, 'b', 'filled');
title('Forced Response y_{forced}(n)')
ylabel('y_{forced}(n)')
xlabel('n')
grid on

figure;
stem(n, y_total, 'filled');
title('Total Response y(n)')
xlabel('n')
ylabel('y(n)'
) grid on
```

## OUTPUT:






## RESULT – AIM 4

- The natural (zero-input) response of the system was obtained using the given initial conditions.
- The forced (zero-state) response due to the input signal was computed correctly.

## CONCLUSION:

This experiment successfully demonstrated the application of the Z-transform in analyzing discrete-time systems. Polynomial multiplication in the Z-domain was verified using convolution, and inverse Z-transform techniques were used to obtain time-domain sequences accurately.
Additionally, the decomposition of the total response into natural and

forced components reinforced the theoretical understanding of LTI system behavior. Overall, the experiment met its objectives and strengthened the practical understanding of Z-transform–based system analysis in digital signal processing.